Justification for Accelerated Publication

Manuscript: Gravity Field Dynamics and Bimetric Gravity: Dual Tetrahedral Field Chambers as the Key to the General Parameters

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This Letter resolves a specific, long-standing open problem in ghost-free bimetric gravity: the theoretical determination of the five free interaction coefficients β_n. Since the Hassan–Rosen framework was established in 2012, these coefficients have been treated as continuous phenomenological parameters. The present work derives them as the unique integer set {1, 0, 16, 0, 1} from the combinatorial geometry of the dual tetrahedral field chamber, using no fitted parameters.

The result is self-contained and algebraic. The full derivation, from the GFD coupling polynomial through the locked bimetric potential, the ghost-freedom proof, the EFT bridge, and the recovery of the IR constitutive law and scalar-tensor action, is presented within the concise format of a Letter. No additional length is required.

Two supplementary files are included. The first is an interactive three-dimensional point-cloud reconstruction of the dual tetrahedral field chamber (fig1_dual_tetrahedral_field_chamber.html), in which four million points sampled uniformly in local frame coordinates are classified by the four half-space inequalities $v_1 = x + y + z$, $v_2 = x - y - z$, $v_3 = -x + y - z$, $v_4 = -x - y + z$. No mesh or curve fitting is used; TetA, TetB, and the shared octahedral Field Origin emerge from the polyhedral boundaries alone. The second is a validation script (run_all_validations.py) that independently verifies every algebraic claim in the manuscript, including the locked parameter set, the coupling polynomial evaluations, the constitutive law recovery, and the kinetic functional.